

Exercise 1 — Parts Order Form Normalization

customerNumber uniquely identifies a customer.
partNumber uniquely identifies a part.
Each part is stored in one cage identified by cageCode.
Employees can serve any customer.
I assume the combination of customerNumber, date, and time uniquely identifies one order.
I assume the same partNumber appears at most once on an order.

UNF

parts_order
customerNumber
customerName
customerType
date
time
employee
partNumber
partName
partType
cageCode
quantityOrdered
unitPrice

1NF

parts_order	
<u>customerNumber</u>	PK
customerName	
customerType	
<u>date</u>	PK
<u>time</u>	PK
employee	
<u>partNumber</u>	PK
partName	
partType	
cageCode	
quantityOrdered	
unitPrice	

The repeating part rows are converted into separate records, so each field contains a single value. The composite key identifies each order line.

2NF

customer
<u>customerNumber</u>
customerName
customerType

part
<u>partNumber</u>
partName
partType
cageCode
unitPrice

parts_order
<u>customerNumber</u>
<u>date</u>
<u>time</u>
employee

order_line
<u>customerNumber</u>
<u>date</u>
<u>time</u>
<u>partNumber</u>
quantityOrdered

Partial dependencies are removed. customerName and customerType depend only on customerNumber, while partName, partType, cageCode, and unitPrice depend only on partNumber. Order-level data is separated from line-item data.

3NF

customer	
<u>customerNumber</u>	PK
customerName	
customerType	

part	
<u>partNumber</u>	PK
partName	
partType	
cageCode	
unitPrice	

parts_order	
<u>customerNumber</u>	PK,FK
<u>date</u>	PK
<u>time</u>	PK
employee	

order_line	
<u>customerNumber</u>	PK,FK
<u>date</u>	PK,FK
<u>time</u>	PK,FK
<u>partNumber</u>	PK,FK
quantityOrdered	

The 2NF relations are already in 3NF because no non-key attribute depends on another non-key attribute.

Exercise 2 — Therapist Appointment Normalization

staffNo uniquely identifies a therapist.
patNo uniquely identifies a patient.
A therapist works at only one branch on a given date.
A patient may have multiple appointments on the same day.
I assume the combination of staffNo, patNo, appointmentDate, and appointmentTime uniquely identifies one appointment.

UNF

appointment	
staffNo	
therapistName	
patNo	
patName	
appointmentDate	
appointmentTime	
branchNo	

1NF

appointment	
<u>staffNo</u>	PK
therapistName	
<u>patNo</u>	PK
patName	
<u>appointmentDate</u>	PK
<u>appointmentTime</u>	PK
branchNo	

2NF

therapist	
<u>staffNo</u>	PK
therapistName	

patient	
<u>patNo</u>	PK
patName	

therapistSchedule	
<u>staffNo</u>	PK
<u>appointmentDate</u>	PK
branchNo	

appointment	
<u>staffNo</u>	PK
<u>patNo</u>	PK
<u>appointmentDate</u>	PK
<u>appointmentTime</u>	PK

Partial dependencies are removed. therapistName depends only on staffNo, patName depends only on patNo, and branchNo depends on staffNo and appointmentDate.

3NF

therapist	
<u>staffNo</u>	PK
therapistName	

patient	
<u>patNo</u>	PK
patName	

therapistSchedule	
<u>staffNo</u>	PK,FK
<u>appointmentDate</u>	PK
branchNo	

appointment	
<u>staffNo</u>	PK,FK
<u>patNo</u>	PK,FK
<u>appointmentDate</u>	PK,FK
<u>appointmentTime</u>	PK

The 2NF relations are already in 3NF because there are no transitive dependencies between non-key attributes.

Exercise 3 — Employee Contract Hours Normalization

eNo uniquely identifies an employee.
contractNo uniquely identifies a contract.
Each contract applies to only one event.
I assume eventNo uniquely identifies an event and its location.
I assume the combination of eNo and contractNo uniquely identifies one employee-hours record for a contract.

UNF

employeeContract
eNo
contractNo
hours
eName
eventNo
eventLoc

1NF

employeeContract	
<u>eNo</u>	<u>PK</u>
<u>contractNo</u>	<u>PK</u>
hours	
eName	
eventNo	
eventLoc	

2NF

employee	
<u>eNo</u>	<u>PK</u>
eName	

contract	
<u>contractNo</u>	<u>PK</u>
eventNo	
eventLoc	

employeeContract	
<u>eNo</u>	<u>PK</u>
<u>contractNo</u>	<u>PK</u>
hours	

3NF

employee	
<u>eNo</u>	<u>PK</u>
eName	

event	
<u>eventNo</u>	<u>PK</u>
eventLoc	

contract	
<u>contractNo</u>	<u>PK</u>
eventNo	FK

employeeContract	
<u>eNo</u>	<u>PK,FK</u>
<u>contractNo</u>	<u>PK,FK</u>
hours	

The transitive dependency is removed. eventLoc depends on eventNo, so event information is separated from the contract relation. Each contract references one event.

Partial dependencies are removed. eName depends only on eNo, while eventNo and eventLoc depend on contractNo. hours depends on the full composite key eNo + contractNo.